

Visible-light promoted synthesis of substituted benzo[c]-chromeno[4,3,2-gh]phenanthridine involving hetero-cross-coupling cyclization and aromatization reaction using organic dye Na₂-eosin Y

A

B. Tech. Project Report Submitted

As Partial Fulfilment of the Requirements for the Degree of

Bachelor of Technology

CH-498

Project I



By

BIJIRGIRI BASUMATARY

Roll No. 200122019

Supervisor: Professor Subhas Chandra Pan

**Department of Chemistry
Indian Institute of Technology Guwahati
Guwahati-781039, Assam
India**

CERTIFICATE

This is to certify that Mr Bijirgiri Basumatary having Roll No. 200122019 has been working in my research group as a project student since 26th August 2023. I am forwarding his project work entitled “*Visible-light promoted metal-free synthesis of 9-isopropylbenzo[c]chromeno[4,3,2-gh]phenanthridine involving hetero-cross-coupling cyclization and aromatization reaction using organic dye Na₂-eosin Y as a Photosensitizer*” for submission of the project report **(Phase-I)**. I certify that he has fulfilled all the requirements according to the rules of this Institute regarding the investigations embodied in his project report and this work has not been submitted elsewhere for a degree.

IIT Guwahati


Prof. Subhas Chandra Pan
(Project Supervisor)



Indian Institute of Technology Guwahati

Guwahati, 781039, India
E. mail : span@iitg.ac.in

Bijirgiri Basumatary

7th Semester B. Tech.

Department of Chemistry

Date: 8th November 2023

STATEMENT

I declare that the work presented in this project titled “*Visible-light promoted metal-free synthesis of 9-isopropylbenzo[c]chromeno[4,3,2-gh]phenanthridine involving hetero-cross-coupling cyclization and aromatization reaction using organic dye Na₂-eosin Y as a Photosensitizer,*” submitted to the Department of Chemistry, and to Professor Subhas Chandra Pan, Indian Institute of Technology Guwahati, for the project report of BTP (Phase-I), is my original work. I have not plagiarized or submitted the same work for the project report.

Bijirgiri Basumatary

200122019

ACKNOWLEDGEMENT

I express my profound gratitude to my research supervisor Professor Subhas Chandra Pan for his guidance, patience, valuable educative discussion and continuous help in all possible ways. Without his guidance and persistent help, this project would not have been possible.

*I also owe a great debt of gratitude to the research scholars of my lab **Mr. Mukesh Kumar** for his constant support, timely suggestions and inspirations with kindness and enthusiasm have enabled me to complete my project.*

Bijirgiri Basumatary

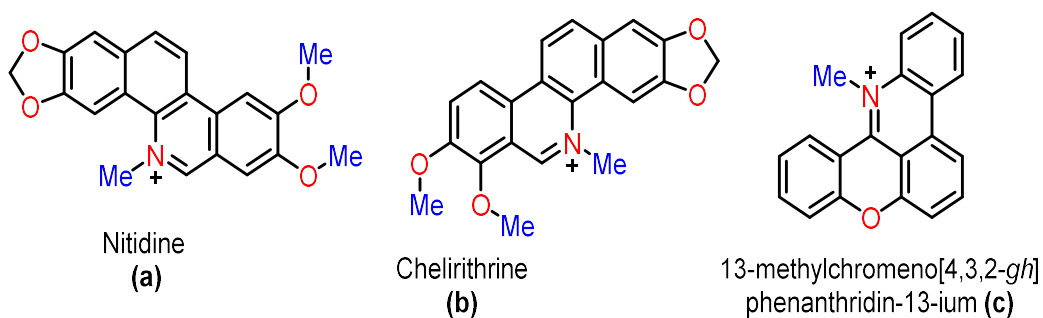
Table of Contents

Contents	Page No.
1.1. Abstract	6
1.2. Introduction.....	6
1.3. Experimental Section	7-9
1.4. Mechanism	10
1.5. Results and Discussion.....	11
1.6. Conclusion	12
1.7. Spectral Analysis.....	12-14
1.8. Reference.....	15

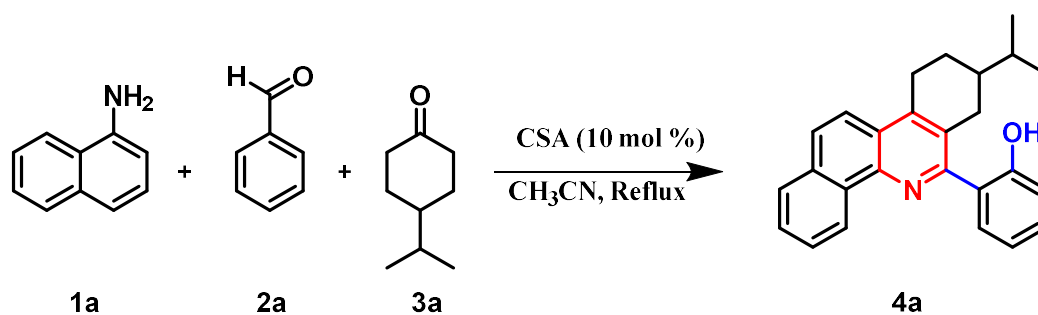
Abstract: Visible–light–induced radical catalyzed synthetic methodology was developed for the synthesis of 9-isopropylbenzo[c]chromeno[4,3,2-gh]phenanthridine starting from 2-(8-isopropyl-7,8,9,10-tetrahydrobenzo[c]phenanthridin-6-yl)phenol in 1,4-dioxane as solvent (Scheme 2), and it will be presented. Derivatives of 9-isopropylbenzo[c]chromeno[4,3,2-gh]phenanthridine were obtained by using the inexpensive organic dye Na₂eosin yellow as a photocatalyst in a substoichiometric amount in the presence of 2×9 W white LEDs visible irradiation. A heterocross-coupling reaction followed by aromatization takes place in a single step without involves any metal catalysts or dehydrogenating agent. Notably 2-(8-isopropyl-7,8,9,10-tetrahydrobenzo[c]phenanthridin-6-yl)phenol derivatives can be easily prepared by multicomponent reaction of commercially available salicylaldehyde, substituted cyclohexanone, and 1-naphthylamine derivatives in a single step.

Introduction: Photoredox-catalyzed reactions have been actively studied in recent years because of such characteristics as simple and mild reaction conditions, low chemical waste generation, and reaction diversity⁵. Particularly in terms of cost-effectiveness and resource availability, organic photoredox catalysts provide benefits. There are several documented organic photoredox catalysts. Among them, Na₂-eosin Y is a commonly accessible non-metallic natural dye that has lately gained extensive usage as a result of its advantages over transition photocatalysts based on metals in terms of both economy and the environment. Although organic synthesis has lately found various uses for visible light photoredox techniques, general interest in the topic dates back considerably further. Photoredox processes don't need stoichiometric chemical oxidants or reductants or radical initiators to take place under mild circumstances, unlike thermal reactions. Due to adjustable photophysical and redox properties, Polycyclic Heteroaromatic Compounds (PAHs) are interesting materials for potential applications in organic electronics. These polycyclic aromatic hydrocarbons (PAHs) are in demand for new materials in diverse scientific and technical fields, such as organic light-emitting diodes (OLEDs), organic field effect (OFELTs), organic photovoltaics (OPVs), solar cells. Therefore, Chemists are still trying to synthesize new molecules having high electronic properties. Benzo[c]phenanthridine alkaloids contain a chrysene skeleton heterocyclic core and are classified as quinoline alkaloids. They are distributed in the plant family and display a broad spectrum of biological activity, including ant microbiology, antifungal and antitumor effects. As flat polyaromatic compounds, they directly

interact with DNA. However, cell cycle arrest and induction of cell death caused by these alkaloids probably occur not only due to DNA damage alone but as a combined result of targeting other cellular structures. Polycyclic Heteroaromatic Compounds having phenanthridine skeletons occur naturally in ring alkaloids, and they have been utilized as traditional or modern medicine or as starting points of drug discovery e.g. Nitidine, Chelirithrine and 13-methylchromeno[4,3,2-*gh*]phenanthridin-13-ium which are shown in below.

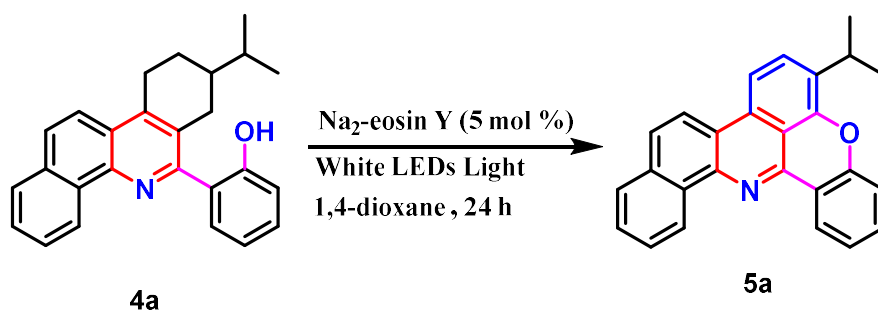


Experimental Section: To an oven-dried 10 mL screw-cap borosilicate vial, 2-(8-isopropyl-7,8,9,10-tetrahydrobenzo[*c*]phenanthridin-6-yl)phenol (0.1 mmol, 32.54 mg), Na₂eosin yellow (5 mol % 3.46 mg) were dissolved in 1,4 dioxane (1.5 mL), and then the reaction mixture was stirred under the irradiation of two 9 W white LEDs bulbs (400 to 700 nm) at a distance of approximately 2-4 cm for 24 h. After completion of the reaction (monitored by TLC analysis), the reaction mixtures were diluted with brine solution (5 mL) and extracted with ethyl acetate (12 mL). The combined organic layer was dried over anhydrous Na₂SO₄ and the solvent was evaporated under reduced pressure. The crude residue thus obtained was purified by column chromatography on silica gel (60-120 mesh) using only hexane as eluent to afford the desired products 9-isopropylbenzo[*c*]chromeno[4,3,2-*gh*]phenanthridine in 86% yield. Each set of a reaction (entry 1-20) was also performed similarly with almost identical yields. Each synthesised compound was structurally confirmed by spectroscopy, including- ¹H NMR, ¹³C NMR, IR, Melting points, X-ray Signal crystal XRD and HRMS spectroscopic technique.

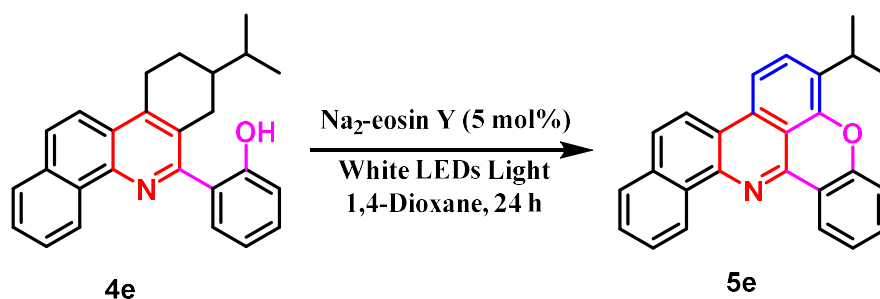


Scheme 1

^aAll reactions were carried out with 1-naphthylamine (**1**, 1 mmol), derivatives of salicylaldehyde (**2**, 1 mmol) and cyclohexanone (**3**, 1 mmol) in 2.5 mL acetonitrile solvent at 81 °C for 5- 9 h.^a isolated yield



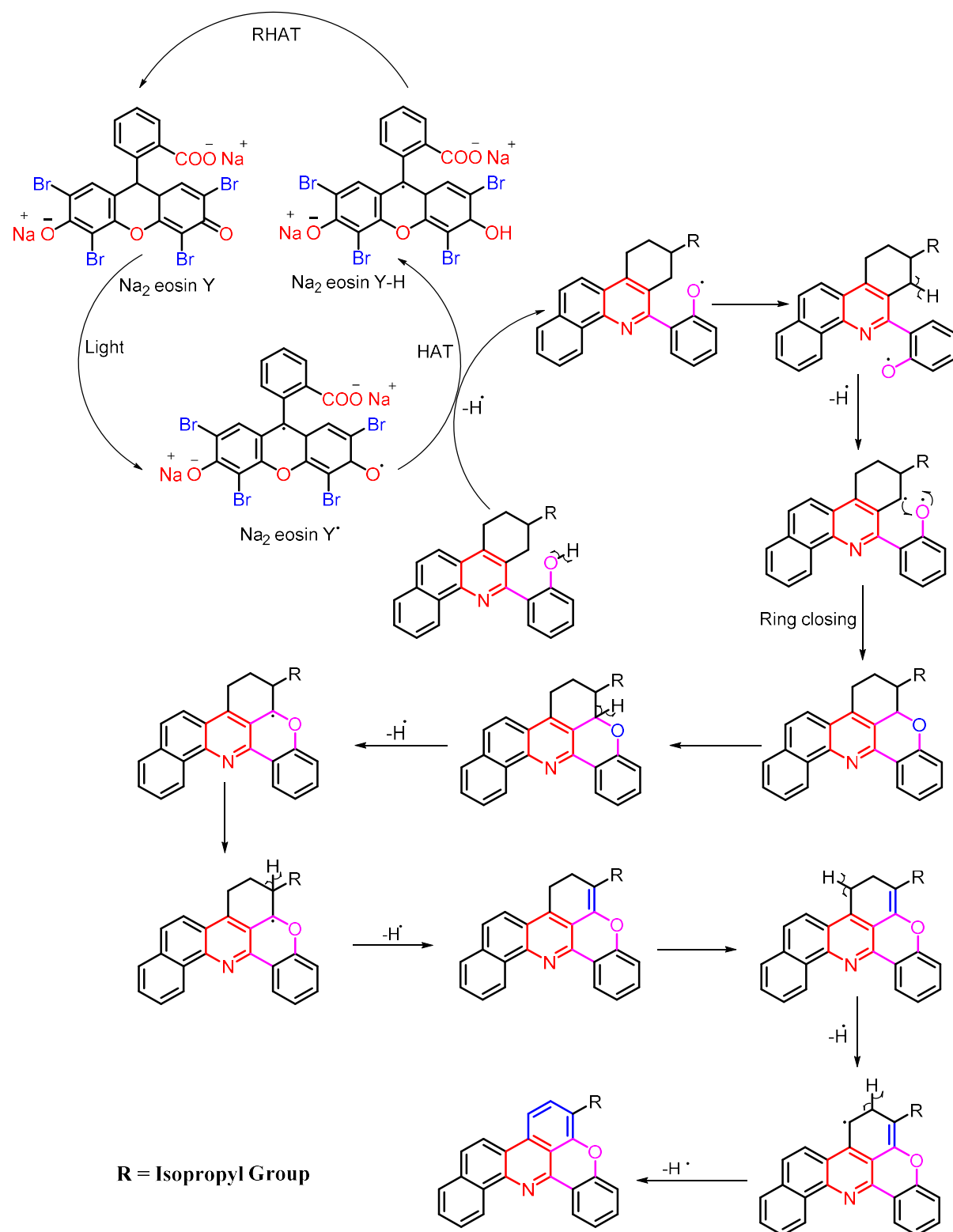
Scheme 2

Table 1 Optimization of the reaction conditions^{a,b}

Entry	Photocatalyst	Visible Light Source	Solvent	Time (Hours)	Yield %
1	Na_2 eosin Y	White LEDs Light (2×9 W)	H_2O	24	ND
2	Na_2 eosin Y	White LEDs Light (2×9 W)	EtOH	24	46
3	Na_2 eosin Y	White LEDs Light(2×9 W)	THF	24	55
4	Na_2 eosin Y	White LEDs Light (2×9 W)	1,4-dioxane	24	86
5	Na_2 eosin Y	White LEDs Light(2×9 W)	1,4-dioxane	18	64
6	Na_2 eosin Y	White LEDs Light(2×9 W)	1,4-dioxane	36	87
7	Na_2 eosin Y	Green LEDs Light (2×9 W)	1,4-dioxane	24	78
8	Na_2 eosin Y	Blue LEDs Light (2×9 W)	1,4-dioxane	24	75
9	Without (Na_2 eosin Y)	White LEDs Light (2×9 W)	1,4-dioxane	24	ND

^aAll reactions were carried out with 2-(8-isopropyl-7,8,9,10-tetrahydrobenzo[c]phenanthridin-6-yl)phenol(0.2 mmol) (**4e**), and Na_2 eosin Yellow (5 mol%) in 1.5 mL 1,4-dioxane solvent and stirred under the irradiation of 2 × 9 W white LEDs bulbs for 24 h. ^bIsolated yield.

Mechanism of Reaction:



Results and Discussions: Primarily, we used the model reaction of 2-(8-isopropyl-7,8,9,10-tetrahydrobenzo[*c*]phenanthridin-6-yl)phenol under visible-light irradiation in the presence of organic dyes like Na₂-eosin yellow and rose Bengal photocatalyst with 1,4-dioxane solvent for 24 hours, maintaining the temperature between 25 and 30 °C as indicated in the optimization table to begin the meaningful study. The reaction time and solvent screening were initially carried out, and the yield was determined through ¹H NMR with area normalisation. Then different organic dyes, such as Na₂eosin Yellow, rose Bengal and rhodamine-B as the photocatalysts (5 mol %) were surveyed in the presence of many solvents such as 1,4- dioxane, THF, EtOH, H₂O, CH₃CN, DMF, Toluene, DCM, etc and also various condition such as in presence of visible light, Room temperature, reflux condition, in presence of sunlight, in dark etc. with oven-dried 10 mL screw-cap borosilicate vial, respectively. Among them, Na₂eosin yellow was the most promising photocatalysts, the suitable solvent was 1,4-dioxane and 2×9 W white LEDs bulbs to delivering the required products in 86% yield. Through using the Na₂ eosin yellow as the photocatalyst, solvents, including 1,4- dioxane, THF, EtOH, DMF, DCM, H₂O and Toluene were screened, respectively. The results show that 1,4-dioxane solvent was superior to others. Remarkably, when the reaction was carried out in an oven-dried 10 mL screw-cap borosilicate vial with Na₂-eosin yellow (5 mol%) and a solvent like 1,4-dioxane 1.5 ml, the yield of 3b was greatly increased to 78%. Then the model reactions were evaluated under 2×9 W white, blue and green LEDs bulbs in oven-dried 10 mL screw-caped borosilicate vial. Further investigation on the amount of photocatalyst and solvent was finally carried out. After intensive experimentation, the optimal conditions were established as follows substituted 2-(7,8,9,10-tetrahydrobenzo[*c*]phenanthridine-6-yl) phenol 1 (0.1 mmol %), Na₂eosin Y (5 mol%), 1,4-dioxane solvent (1.5 mL) with the irradiation of 2×9 White LEDs bulbs for 24 hour.

Conclusion: A lot of interest has already been shown in visible light photo redox catalysts with metal free as tools for organic synthesis transformations. Na₂-eosin Y is a desirable substitute for redox active metal-free in a number of applications; in certain instances, it even works better than them. Na₂-eosin Y photocatalyst has been used to produce reactive intermediates such as aryl radical, iminium ion, enone radical anion, hetero cyclisation, aromatisation amine functionalisationhydroxylation, reduction and oxidation reactionetc. Because it is non-toxic, easy to handle, less expensive, and environmentally friendly, the photocatalyst Na₂-eosin Y has received a lot more attention. To generate an enantioselective reaction, Na₂-eosin Y catalysis has also been combined with other catalyst modes, such as enamine catalyst and hydrogen bond promoted catalyst. It has been reported how Na₂-eosin Y photocatalysis is used in continual new technology. Overall, Na₂eosin Y is an attractive and environmentally friendly photocatalyst for organic synthesis due to its high availability, substantial absorption in the visible portion of the spectrum, and appropriate redox potential value.

Spectral Analysis:

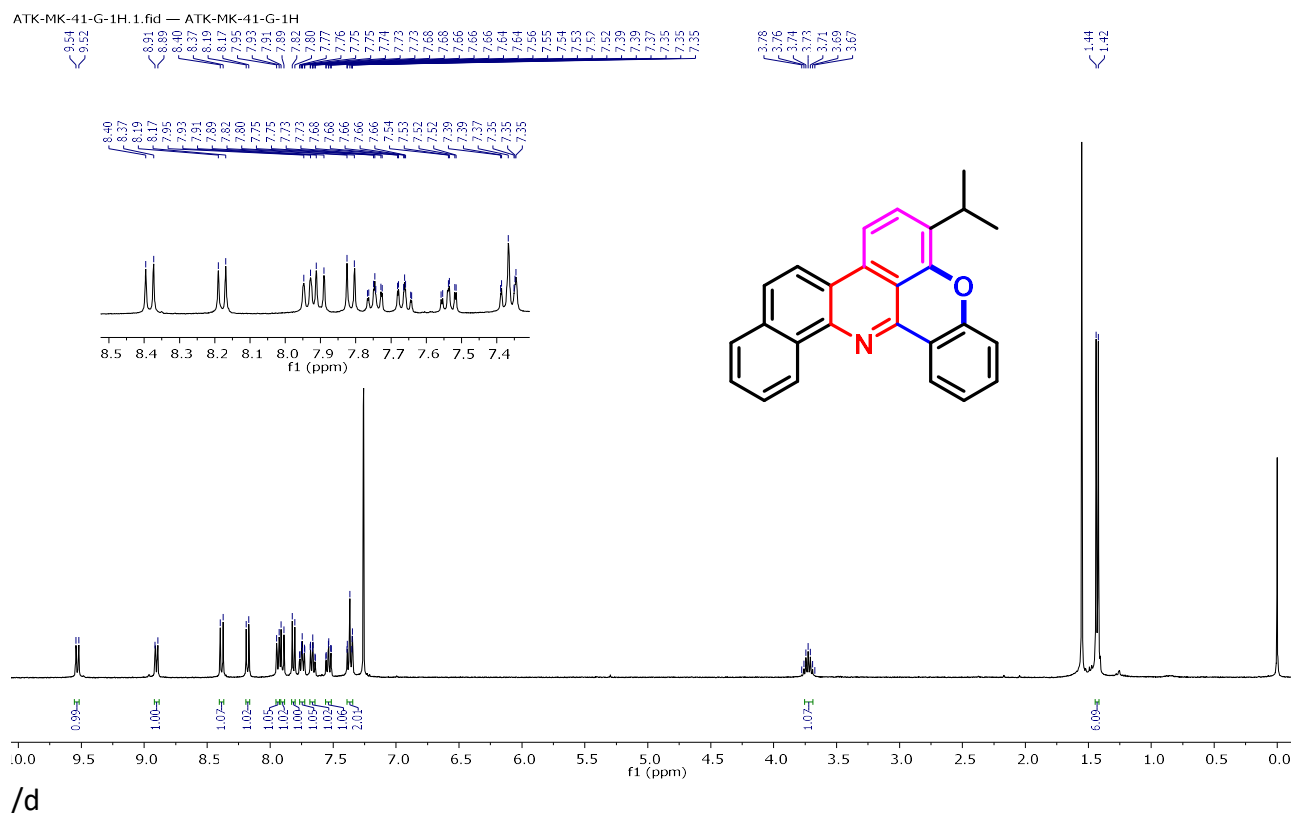


Figure 2 ¹H NMR spectra

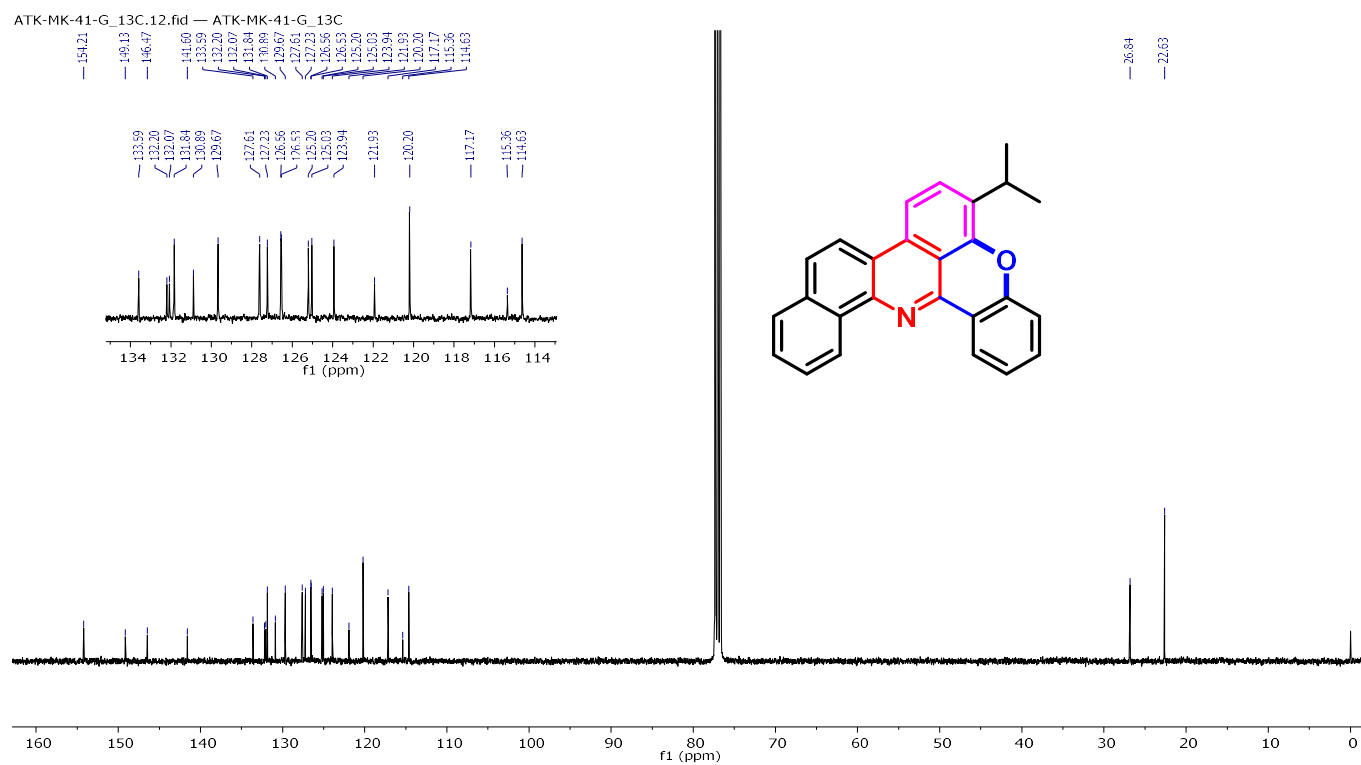
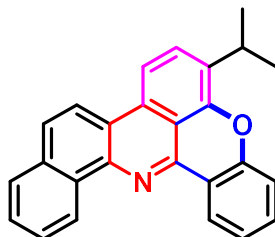
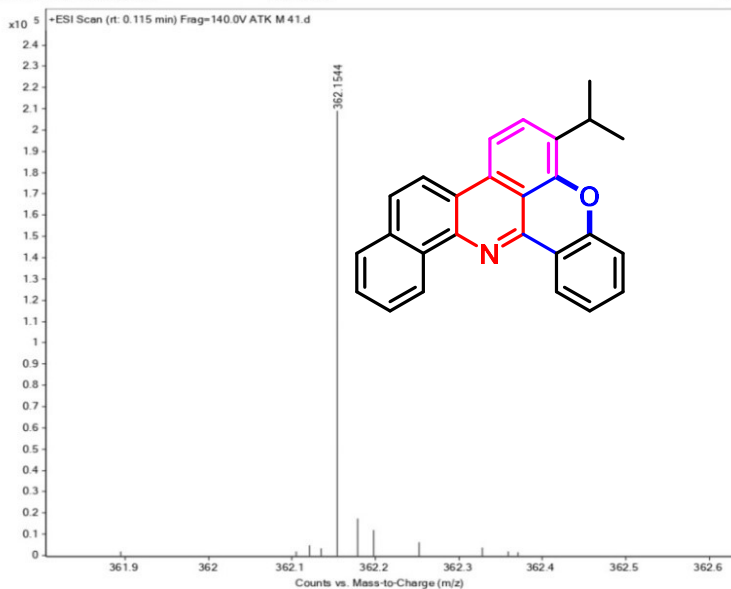


Figure 3 ^{13}C NMR spectra

Sample Name	Sample28	Position	P1-C6	Instrument Name	QTOF
User Name	SYSTEM (SYSTEM)	Inj Vol	5	InjPosition	
Sample Type	Sample	IRM Calibration Status	Success	Data Filename	ATK M 41.d
ACQ Method	DIRECT MASS_POSITIVE_01.m	Comment		Acquired Time	03-04-2023 16:26:12 (UTC+05:30)



9-isopropylbenzo[c]chromeno[4,3,2-g]phenanthridine

Yellow solid (84% yield)., ^1H NMR (400 MHz, Chloroform-*d*) δ 9.53 (d, J = 8.3 Hz, 1H), 8.90 (d, J = 9.0 Hz, 1H), 8.38 (d, J = 9.0 Hz, 1H), 8.18 (d, J = 8.5 Hz, 1H), 7.94 (d, J = 7.9 Hz, 1H), 7.90 (d, J = 8.8 Hz, 1H), 7.81 (d, J = 8.5 Hz, 1H), 7.77 - 7.73 (m, 1H), 7.68 - 7.64 (m, 1H), 7.54 (td, J = 8.0, 1.7 Hz, 1H), 7.39 - 7.34 (m, 2H), 3.75 - 3.69 (m, 1H), 1.43 (d, J = 6.9 Hz, 6H)., ^{13}C NMR (101 MHz, Chloroform-*d*) δ 131.84, 129.67, 127.61, 127.23, 126.56, 126.53, 125.20, 125.03, 123.94, 120.20, 114.63, 77.33, 26.84., IR (KBr) $\nu_{\text{max/cm}^{-1}}$ 2959, 1740, 16909, 1454, 1154, 1037, 753.

References:

1. M. Tanioka, A. Kuromiya, R. Ueda, T. Obata, A. Muranaka, M. Uchiyama and S. Kamino, *Chem. Commun.*, 2022, **58**, 7825-7828.
2. F. Mohamadpour, *ACS Omega*, 2022, **7**, 8429-8436.
3. D. P. Hari and B. König, *Chem. Commun.*, 2014, **50**, 6688-6699.
4. A. Bosveli, T. Montagnon, D. Kalaitzakis and G. Vassilikogiannakis, *Org. Biomol. Chem.*, 2021, **19**, 3303-3317.
5. Y. Song, H. Zhang, J. Guo, Y. Shao, Y. Ding, L. Zhu and X. Yao, *Eur. J. Org. Chem.*, 2021, **2021**, 5914-5921.
6. F. Mohamadpour, *Front. Chem.*, 2022, **10**, 880257
7. R. Li, X. Chen, S. Wei, K. Sun, L. Fan, Y. Liu, L. Qu, Y. Zhao and B. Yu, *Adv. Synth. Catal.*, 2018, **360**, 4807-4813.
8. G. Shukla, A. Dahiya, T. Alam and B. K. Patel, *Asian J. Org. Chem.*, 2019, **8**, 2243-2248.
9. A. K. Sahoo, A. Rakshit, A. Dahiya, A. Pan and B. K. Patel, *Org. Lett.*, 2022, **24**, 1918-1923.
10. Y. Liu, P. Zhou, Y. Xu, Z. Yang and D. Wang, *Chem. Commun.*, 2023, **59**, 1681-1684.
11. M. Jarrahi, B. Maleki and R. Tayebee, *RSC Adv.*, 2022, **12**, 28886–28901.
12. S. Yashmin, R. Ali, S. Mondal and A. T. Khan, *Chem. Commun.*, 2022, **58**, 5853-5856.
13. S. Yashmin, S. Mondal, R. Das, P. Banerjee and A. T. Khan, *Org. Biomol. Chem.*, 2022, **20**, 7302-7315.
14. R. Gattu, P. R. Bagdi, R. S. Basha and A. T. Khan, *J. Org. Chem.*, 2017, **82**, 12416-12429.

15. A. Borissov, Y. K. Maurya, L. Moshniaha, W.-S. Wong, M. Żyła-Karwowska and M. Stępień, *Chem. Rev.*, 2022, **122**, 565-788.
16. A. Ambrosi, C. K. Chua, A. Bonanni and M. Pumera, *Chem. Rev.*, 2014, **114**, 7150-7188.
17. C. G. Neochoritis, T. Zarganes-Tzitzikas, K. Katsampoxaki-Hodgetts and A. Dömling, *J. Chem. Educ.*, 2020, **97**, 3739-3745.